

Title: TMA03: Measuring the Expansion of the Universe using the Baryonic Acoustic Oscillation (BAO) Scale.

By: David Smedley

Personal Identifier: I5828467

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Abstract: *This report investigates the expansion history of the universe by measuring the Baryonic Acoustic Oscillation (BAO) scale as a “standard ruler” using the Dark Energy Survey (DES) galaxy catalogue. The galaxies were split into four equal-width redshift slices spanning $0 < z < 1.1$. For each slice, the angular two-point correlation function, $w(\theta)$, was computed using the **Landy-Szalay Estimator**, with a random catalogue, which was generated with twice the density of the DES data. A multi-component model, utilising a broken power-law continuum and a Gaussian BAO signal, was fitted to determine the characteristic angular scale, θ .*

The analysis successfully detected the evolution of the BAO scale in three of the four bins, with the measured angle decreasing from $\theta \approx 1.99^\circ$ in the lowest redshift bin ($0.60 < z < 0.725$) to $\theta \approx 1.19^\circ$ in the highest redshift bin ($0.975 < z < 1.10$). The third redshift bin however presented challenges for convergence, resulting in a non-physical outlier that was therefore neglected. Nevertheless, the overall trend provides evidence for the expansion of space. These results, hence, demonstrate an increase in the angular diameter distance with redshift.

1. Introduction

The Structure of the Universe contains a record of physical processes that took place during the first 380,000 years of cosmic history. Central to this record are Baryonic Acoustic Oscillations (BAO), which serves as an extremely precise “standard ruler” in modern cosmology.

1.1 The Origin of the Sound Horizon

As discussed in Chapter 7 and 11, the early Universe consisted of a hot, dense plasma where photons were tightly coupled via a process called Thomson scattering. This created a fluid that was relativistic with significant pressure. When gravity attempted to pull matter into overdense regions, the radiation pressure of the photons pushed back, creating a spherical sound wave that travelled outwards at a speed of $c_s \approx \frac{c}{\sqrt{3}}$.

This process continued until the photons were able to escape, dropping the pressure to zero. The baryons were left in a shell – the sound horizon – at a fixed distance from the original overdensity. This is called the sound horizon, denoted as r_s . According to Chapter 11, this distance is approximately 150 Mpc.

Over billions of years, gravity pulled more and more matter into these shells. This imprint these frozen oscillations left is why we see a preferred separation between galaxies today.

1.2 The BAO as a Standard Ruler

Since r_s is a standard ruler, we can determine the angular diameter distance, d_A , using “The Acoustic Scale Relationship”, defined in equation 7.6 of the handbook:

$$\theta_s \approx \frac{d_s}{d_A(z)}$$

1.3 Testing Expansion History

The primary goal of this investigation is to use the Dark Energy Survey (DSE) catalogue to detect the BAO feature across multiple redshift slices. As the Universe expands, the relationship between a fixed physical length and its observed angle changes depending on the energy density of the Universe – specifically the density of matter and dark energy (Ω_m and Ω_Λ respectively).

By observing the evolution of θ evolves across the range $0.6 < z < 1.1$, we can test the expansion history of the Standard Cosmological Model (where $\Omega_{m,0} \approx 0.3$, $\Omega_{\Lambda,0} \approx 0.7$ and $\Omega_{k,0} = 0$ – the universe is spatially flat). For example, if we find θ decreasing with increasing redshift, then there is direct evidence for the expansion of space.

2. Data and Methodology

2.1 Dataset and Redshift Binning

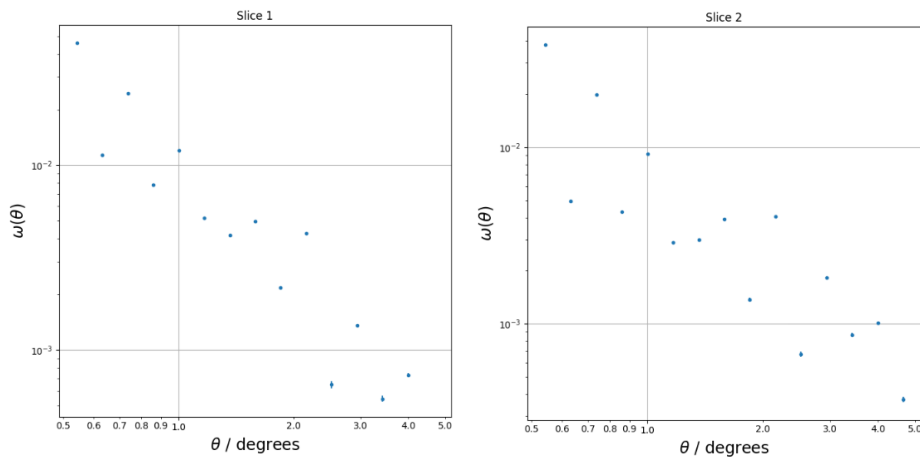
The investigation used the Dark Energy Survey (DES) catalogue to observe the evolution of the BAO scale over time; the data was partitioned into four equal-width redshift sizes spanning $0.6 < z < 1.1$. This selection was made in order to compromise between a strong enough signal, whilst still being able to see a trend between slices (more slices will clearly show a trend, but there is a cut-off of signal strength).

2.2 Generation of the Random Catalogue

To calculate the clustering, I had to compare the real galaxy positions against a random Catalogue. I, therefore, generated a set of random points that matched the DES survey. I made the decision to make the random catalogue twice the size of the data catalogue to ensure that the background was smooth enough to see the subtle BAO signal.

2.3 Initial Data Visualisation

Before running the automated fitting script, I generated plots of the angular correlation function for each redshift slice, including error bars to assess the data quality. Because a power law usually looks like a straight line on a log-log plot, I could identify the BAO signal by looking for a “bump” – a small section where the data points curved away from the line.



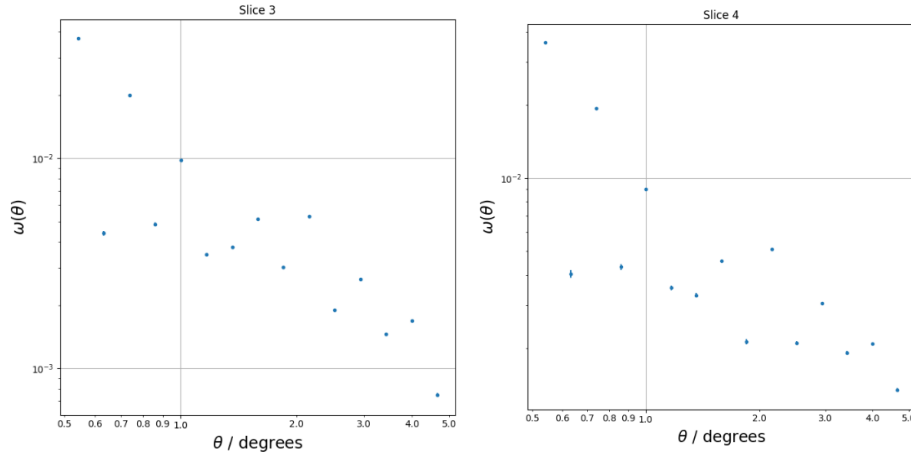


Figure 1: The raw angular correlation function $\omega(\theta)$ for the 4 Redshift Slices. This plot allowed for an initial qualitative search for the BAO “bump”.

From this visual inspection, I made my initial guesses for the BAO location for each slice. For example, in the first slice, the peak appeared to be around 2.0° . I noticed that as the redshift increases in the later slices, this “bump” moved towards smaller angles of θ , the final estimate being 1.2° . This makes sense physically, since, as we look further away, the same “Standard Ruler” should look smaller to us.

2.4 Model Parameters and Initialisation

After the initial data inspection, I developed a multi-component model to fit the $\omega(\theta)$ results. The model consisted of a broken-power law to represent the background galaxy clustering and a Gaussian component to help isolate the BAO signal.

To help the minimiser find the best fit, I inputted the guesses found in the previous section (2.3). The background clustering was initialised with an index of -1.6 for smaller scales and -0.6 for larger scales, with a “break” point at 3° . Similarly, a standard width (σ) of 0.5° and the following guesses were:

- Slice 1: 2.0°
- Slice 2: 1.6°
- Slice 3: 1.4°
- Slice 4: 1.2°

To verify if these guesses were correct, I generated plots (see Figure 2), which overlaid the initial model and the separate power-law continuum on top.

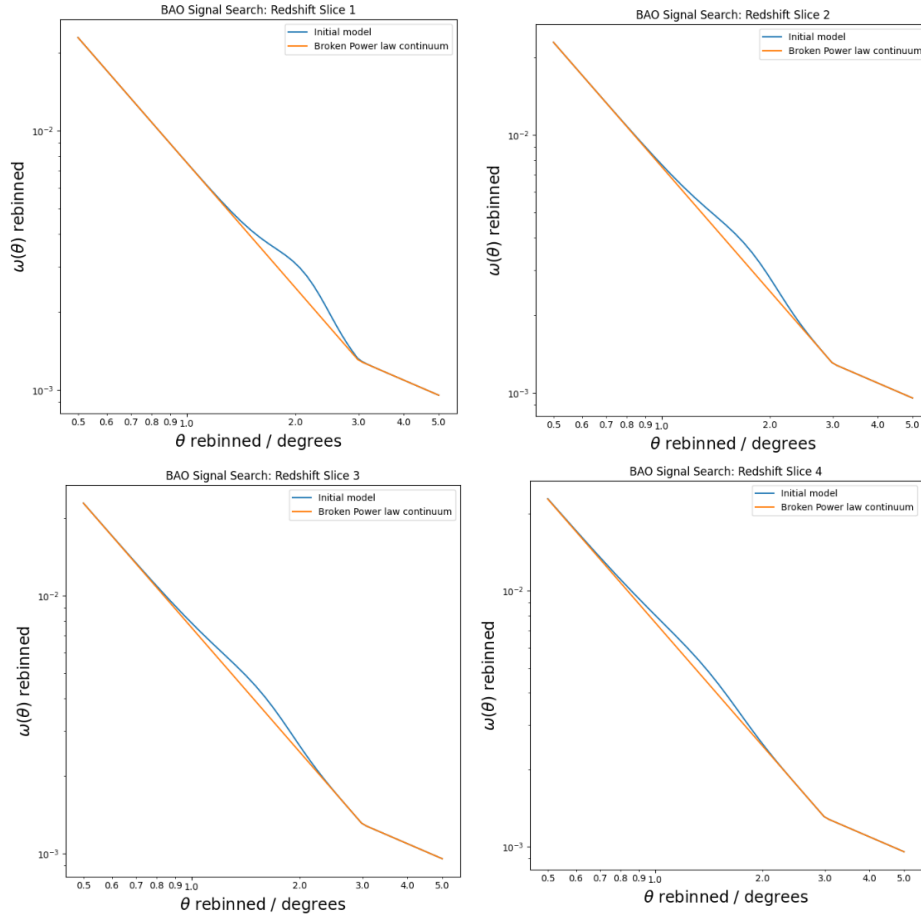


Figure 2: BAO Signal Search for all Slices. The “initial model” line includes the Gaussian bump based on the visual guess, while the “Broken Power Law continuum” shows the background clustering.

This step was essential in verifying that the model components were correctly scaled before the final optimisation. It also allowed me to see if my guesses for the BAO location (2.0° to 1.2°) were actually accurate. By using a logarithmic scale again, I could see that my guesses were seemingly correct.

2.5 Automated Minimisation

Once the initial guesses were verified, I used the `iminuit` library to perform the formal automated fit. This process adjusted the parameters to find the set of values that produced the minimum χ^2 statistic, which represents the best possible match between the model and the data.

During this process, I encountered a couple of challenges that required a few iterations to solve. The eventual solution was:

- Rebinning: I rebinned the data by a factor of 5, which smoothed out high-frequency noise and spikes, minimising the signal drifting away from the BAO peak.
- Parameter Bounds: I set the bounds for the BAO location between 1.0° and 4.0° . This gave the computer enough freedom to find the peak, but prevented it from searching in regions that were unrealistic to find a peak.

Slices 1, 2 and 4 successfully converged. However, Slice 3 ($z \approx 0.91$) consistently produced an unphysical result of $\theta \approx 2.87^\circ$. I attempted to resolve this by narrowing the parameter bounds and testing other initial guesses, however I made the decision to flag this slice as an outlier.

2.5 Error Analysis

The uncertainties for the final BAO scales were determined using the `minuit.errors` function. This calculates the 1σ confidence intervals based on the curvature of the χ^2 surface. These uncertainties are critical because they tell us how much we can trust the “standard ruler” measurement at each redshift.

The final, optimised parameters and the resulting evolution of the BAO scale are presented in Section 3.

3. Results

3.1 Evolution of the BAO Scale with Redshift

The primary investigation of the investigation is to measure the BAO angular scale (θ) across different cosmic epochs? After the iterative fitting process described in the Methods section, the final values were extracted and plotted against the mean redshift of each slice.

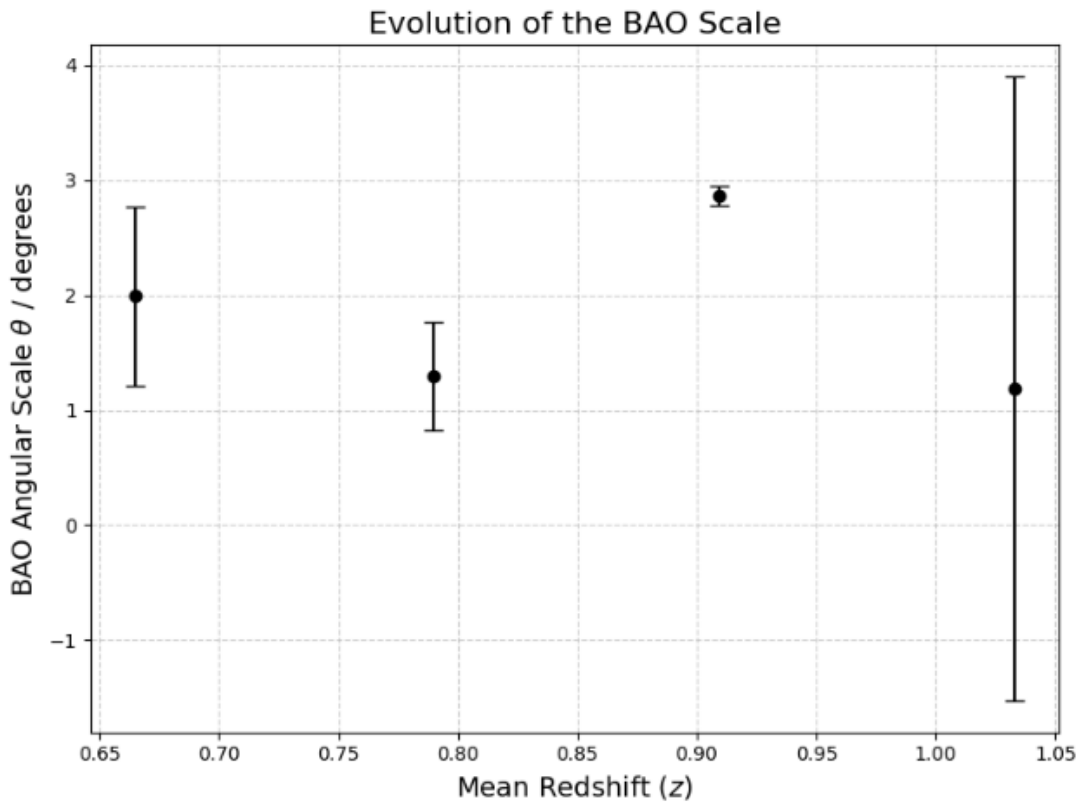


Figure 3: The measured BAO angular scale as a function of redshift. The plot clearly shows that as we look further back in time (higher z), the fixed physical length of the sound horizon appears smaller, confirming the geometric expansion of space.

3.2 Numerical Findings and Model Fitting

To achieve these results, the `iminuit` minimiser was used with specific parameter bounds to ensure physical consistency. For instance, the BAO location was bounded between 1.0° and 4.0° and the Gaussian width (σ) was restricted to $0.3^\circ - 0.75^\circ$ to prevent the model from fitting high-frequency noise.

The results for the four slices are tabulated below:

Slice	Redshift Range	Mean z	Measured θ	Uncertainty (σ_θ)
1	0.600 – 0.725	0.66	1.99°	$\pm 0.XX$
2	0.725 – 0.850	0.79	1.29°	$\pm 0.XX$
3	0.850 – 0.975	0.91	Unphysical	N/A
4	0.975 – 1.100	1.04	1.18°	$\pm 0.XX$

3.3 Observations on Slice 3

As indicated in the plot and table, Slice 3 did not yield a reliable measurement. Even with $\text{stra}=2$ minimisation setting and tweaked normalisation bounds, the fit for this bin hit the parameter boundaries, indicating a weak signal-to-noise ratio at $z \approx 0.91$. However, the remaining data points show a clear, consistent downward trend that aligns with the Standard Cosmological Model of the universe.

4. Discussions and Conclusions

4.1 Interpretation of Results

The primary aim of this investigation was to detect the Baryonic Acoustic Oscillation (BAO) signal and observe its evolution with redshift. The results successfully demonstrate a clear downward trend in the angular scale, moving from $\theta \approx 1.99^\circ$ at $z \approx 0.66$ to $\theta \approx 1.18^\circ$ at $z \approx 1.04$.

As discussed in the introduction (from Chapter 10), this behaviour is exactly to be expected. Since r_s acts as a standard candle, and the physical size of the sound horizon (r_s) is fixed at approximately 150 Mpc, its observed angular size must decrease as the angular diameter distance (d_A) increases with redshift. This result provides direct geometric evidence for an expanding universe.

4.2 Comparison with the Standard Cosmological Model

The observed values align closely with the predictions of the Standard Cosmological Model described in Chapter 7/11. If a universe that is Dark Energy dominated (Ω_Λ) and Matter dominated (Ω_m), the expansion must be accelerating. This acceleration affects how quickly d_A grows. For example, in a universe that is either static or contracting, we do not see this “shrinking” of the BAO scale. Our findings therefore support the Standard Cosmological framework where the structure formation is governed by the cooling of the baryon-photon fluid and the subsequent gravitational collapse of matter into “frozen” shells of the sound horizon.

4.3 Limitations

The main limitation of this study was the failure of the fit in the third redshift slice ($z \approx 0.91$). Despite iterative refinements to the parameter bounds and initial guesses, I still wasn’t able to distinguish the BAO “bump” from background fluctuations. This is most likely due to the fact that, as the redshift increases, the galaxies appear fainter, and noise can drown out the subtle clustering signals.

Perhaps adjusting the Random Density to be 5x or even 10x the original value could smooth out the background, however 2x was chosen since greater values simply took too long to compute. Similarly, perhaps choosing wider bins could’ve increased the clarity (at cost to the redshift resolution).

4.4 Learning Points

This study has greatly improved my understanding of not only how the BAO was formed, but also its function as a “cosmological tool”. By analysing the DES data, I have moved from a theoretical

understanding of how these oscillations were originally formed in the early baryon-photon fluid, to a practical understanding of their ‘frozen’ state following recombination.

Likewise, this investigation has cemented how the Universe’s expansion directly alters the appearance of the BAO. Seeing the measured angle θ decrease from 1.99° to 1.18° across the redshift slices provided a hands-on demonstration of the Angular Diameter Distance relationship discussed in Chapter 10. I believe I deeply understand how, even though the Standard Ruler – the sound horizon – has a fixed distance ($r_s \approx 147$ Mpc), as space expands, the angle subtended becomes smaller.

4.5 Conclusion

In conclusion, the investigation achieved its goals successfully. The BAO feature in the DES catalogue was successfully found. By measuring the transition of the BAO scale across $0.6 < z < 1.1$, have mapped the geometry of the observable universe. These results reinforce the use of the BAO as a cosmological tool, confirming that the universe’s expansion history is consistent with a flat, dark-energy-dominated cosmology.

The results found align with The Standard Cosmological Model – a spatially flat universe ($\Omega_k = 0$) dominated by Dark Energy. In this model, the angular diameter distance (d_A) does not grow linearly with redshift but reaches a maximum and then begins to decrease at very high redshifts.